Solution class Participation 3

Block size = 1024 bytes , Record size = 32 bytes

Block factor of DB Block = 1024 / 32 = 32 Records / Block

Block factor of Index Block = 1024 / 16 = 64 Record / Block

Sparse Index on key = 1 Record from Each DB Block
= Total DB Blocks

Total no. of DB Blocks = 16384 / 32 = 512 Blocks.

No. of entries at First level Index = No. of DB Blocks
= 512 entries

no. of 1st level Index Blocks = 512 / 64 = 8 Blocks

no. of 2nd level Index Block = 1 Block (8 entries only)

Total no. of Index levels = 2

Total no. of Index Blocks = 8 + 1 = 9

Total no. of Blocks at 2nd level = 1

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Class participation ~~B~~

Q.) Block size = 1024 Bytes
Key size = 15 bytes
Pointer = 5 Bytes

Total no. of Records = 500000
Record size = 200 Bytes

Block Factor of DB Block = 1024/200 ≃ 5 Records per Blo

Fanout of B tree

$n \times key + (n+1) \times Pointer \leq 1024$

$15n + (n+1) \times 5 \leq 1024$

$20n \leq 1019$

$n \leq 1019/20 \cong 50$

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Q.2 5 Record / Block = 10 key + 11 Pointers

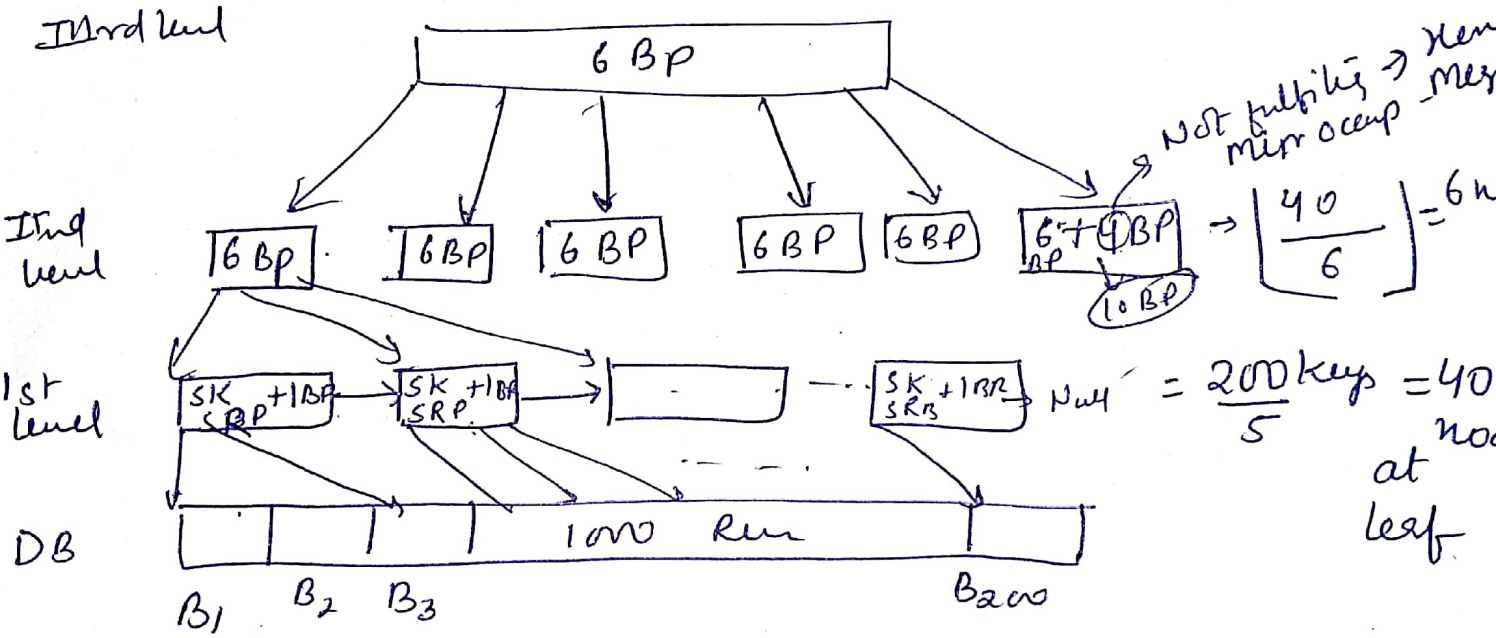
B tree Index Block

min occ = 6 BP + 5 key

Max. occ = 10 key + 11 pointer

Total Database Block = 1000/5 = 200 Blocks

Largest no. of Blocks = Min occupancy in Each index Block.
(sparse B+ tree)



$\left\lceil \frac{40}{6} \right\rceil = 6$ n

$= \frac{200 \text{ keys}}{5} = 40$ nodes at leaf

To obey the minimum occupancy, we go through the floor value because if we take ceiling value, i.e. 7 node; the 7th node only contains 4 BP which is less than min occupancy.

Thus Total no. of nodes

→ 200 + 40 + 6 + 1 = 247

DB, 1st level, 2nd lev, Root lev

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